

RW-0.03 Interface Control Document

Small Reaction Wheel - 0.03 Nms

Hardware Revision 8

Document Number:	AC-ICD-00108
Version:	2.2
Date:	2025-07-22
Classification:	Confidential
Product:	RW-0.03 Small Reaction Wheel
Hardware Revision:	8
Prepared By:	AnyCompany Engineering Team
Approved By:	Systems Engineering

AnyCompany

1. Document History

This section records all revisions to this Interface Control Document.

Version	Date	Author	Description
1.0	2014-03-15	J. Morrison	Initial release for RW-0.03 production baseline
1.5	2018-06-10	M. Chen	Updated sign convention (+CCW from top); added watchdog timer (500ms)
2.0	2022-01-20	R. Tanaka	Updated to NSP v2 protocol; revised CRC to CRC-CCITT polynomial
2.2	2025-07-22	Engineering Team	Revised fault handling; updated diagnostic channels; corrected EDAC map

1.1 Applicable Documents

ID	Document Number	Title
RD-1	AC-DS-00108	RW-0.03 Product Data Sheet v1.0
RD-2	AC-UM-00108	RW-0.03 User Manual
RD-3	NASA GSFC-STD-7000A	General Environmental Verification Standard (GEVS)
RD-4	ECSS-Q-ST-70-02C	Thermal Vacuum Outgassing Test for Spacecraft Materials
RD-5	RFC 1055	Serial Line Internet Protocol (SLIP)
RD-6	IEC 61000-4-2	Electrostatic Discharge Immunity Test

1.2 Distribution List

Recipient	Organization	Copy Type
Systems Engineering	AnyCompany	Controlled
Flight Software Team	AnyCompany	Controlled
Quality Assurance	AnyCompany	Controlled
Customer Integration	External	Controlled

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3. Scope and Applicability

This Interface Control Document (ICD) defines the mechanical, electrical, environmental, and software interfaces for the AnyCompany RW-0.03 small reaction wheel, hardware revision 8. It establishes the requirements and constraints for proper integration and operation.

The RW-0.03 is a single-axis reaction wheel providing **0.03 Nms** angular momentum storage with **1.5 mNm** peak torque. Designed for attitude determination and control subsystems (ADCS) on CubeSats and small satellites in the 3U to 12U class. The RW-0.03 shares a common design family with the RW-0.01, featuring identical electronics but a taller flywheel assembly (40 mm vs 29.5 mm height) for increased momentum storage.

Flight Heritage: 90 units with 510+ cumulative flight years since 2008. This is the most mature product in the AnyCompany small reaction wheel line, with 17+ years of continuous operational heritage. Common design architecture with the RW-0.01.

This document is applicable to:

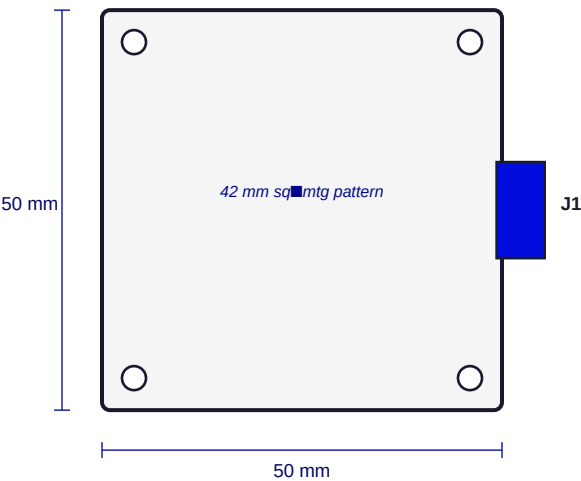
- RW-0.03 hardware revision 8
- Firmware version 2.x and later
- All production variants (standard and extended temperature)

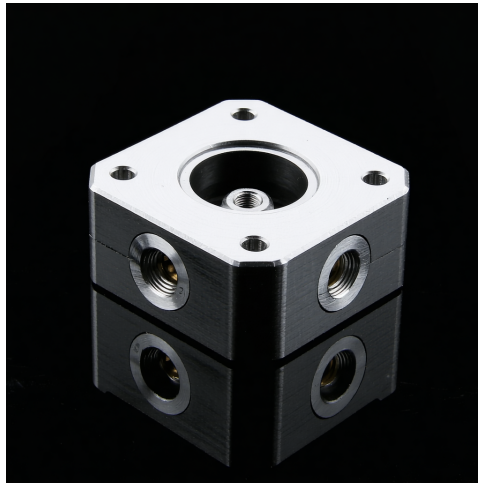
4. Mechanical Interface

4.1 Physical Specifications

Parameter	Value	Units	Notes
Mass	< 185	g	Including connector, excluding harness
Dimensions (L x W x H)	50 x 50 x 40	mm	Taller profile than RW-0.01 (29.5 mm)
Flywheel Diameter	44	mm	Internal rotor assembly
Mounting Pattern	42 x 42	mm	Square, 4x M3 corner holes
Mounting Fasteners	M3		4x corner positions, 6 mm min thread depth
Momentum Storage	0.03	Nms	At maximum rated speed
Maximum Torque	1.5	mNm	Peak, all operating conditions
Control Modes	Speed, Torque		Selectable via SET_MODE command

Figure 3: Mechanical Outline Drawing (Top View)





RW-0.03 reaction wheel assembly (product photograph)

4.2 Mounting Interface

The wheel uses four M3 mounting holes in a 42 mm square pattern at the housing corners. A minimum boss or spacer height of **4 mm** is required beneath all four mounting points to prevent warping of the housing when fasteners are tightened. A precision-machined boss is the recommended approach. Mounting torque: 0.4 to 0.6 Nm with thread-locking compound (Loctite 222 or equivalent).

The mounting interface must provide a flat, clean surface. The wheel frame must not be warped when mounted, as this can impede proper function of the bearings and flywheel. Ensure no debris enters the bearing cavity during integration.

4.3 Remove Before Flight Items

Item	Action	Notes
Kapton bearing covers (2x)	May remove	Protects bearings from particulate contamination during ground handling
Connector protective cap	Remove	Red anodized cap on J1 connector; prevents ESD and contamination
Desiccant packet	Remove	Internal humidity control for storage; must be removed before flight
Shipping restraint	Remove	Foam insert securing flywheel during transport

5. Environmental Requirements

5.1 Storage

Store in a clean environment (ISO Class 8 or better) to prevent bearing contamination. Humidity below 60% RH to prevent rotor corrosion. The wheel may be stored in a sealed antistatic bag with desiccant. Storage temperature range: -40 to +85 degrees C. Maximum storage duration without re-inspection: 24 months.

5.2 Thermal

Parameter	Min	Typ	Max	Units	Notes
Survival Temperature	-40		+100	C	Unpowered, at baseplate interface
Operating Temperature	-30	+20	+70	C	At mounting interface, powered
Qualification Range	-35		+75	C	Derated from qual for flight margin
Cold Start Temperature	-30		+70	C	Safe to apply power
Max Temperature Rate			5	C/min	During powered operation
Thermal Dissipation		0.2	1.2	W	Speed-dependent, see datasheet

Temperature is defined as the temperature at the baseplate mounting interface in the nominal configuration. Operating temperature range has been derated relative to qualification range. Contact AnyCompany for extended temperature variant information.

5.3 Pressure and Outgassing

The wheel operates in sea-level atmosphere and in hard vacuum. It has not been qualified to operate at high-altitude transitional pressures, and should not be powered during launch ascent unless additional testing confirms no arcing risk. All materials meet standard outgassing requirements: **TML < 1%**, **CVCM < 0.1%** per ECSS-Q-ST-70-02C.

5.4 Vibration

Frequency Range	Level	Slope	Duration
20 to 80 Hz	0.008 to 0.04 g ² /Hz	+6 dB/oct	2 min/axis
80 to 500 Hz	0.04 g ² /Hz	Flat	2 min/axis
500 to 2000 Hz	0.04 to 0.004 g ² /Hz	-6 dB/oct	2 min/axis
Overall (Qual)	12.87 Grms		3-axis sequential
Overall (Accept)	9.10 Grms		3-axis sequential (-3 dB)

Qualified to NASA GEVS (GSFC-STD-7000A) random vibration levels. Contact AnyCompany for qualification to more severe environments or sine vibration requirements.

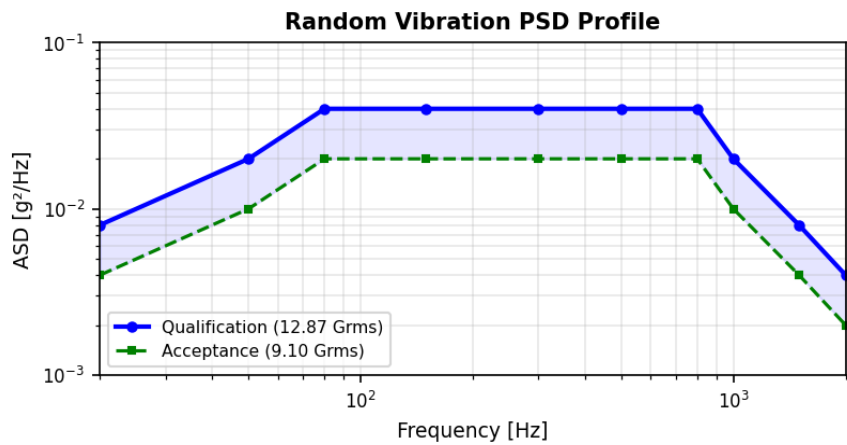


Figure 4: Random Vibration PSD Profile (Qualification and Acceptance Levels)

5.5 Radiation

Total Ionizing Dose (TID): **24 krad** at board level behind 2 mm aluminum equivalent shielding. Single-Event Latchup (SEL) immunity verified to LET > 40 MeV-cm²/mg. Single-Event Upset (SEU) rate: less than 1 per year in typical LEO environment. Components are radiation-lot tested per MIL-STD-883 Method 1019.

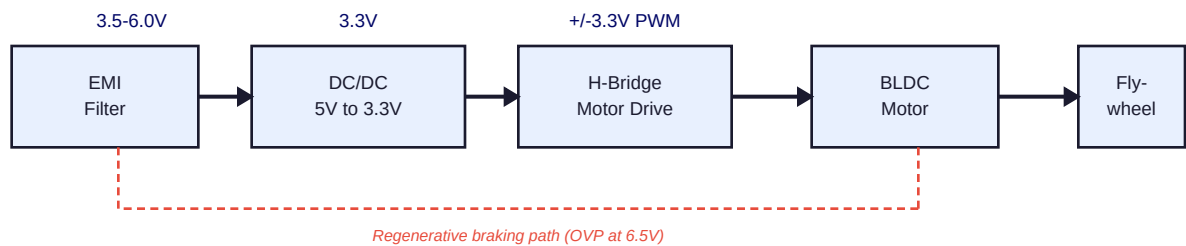
Parameter	Value	Notes
TID Tolerance	24 krad	Board level, 2mm Al equivalent shielding
SEL Threshold	LET > 40 MeV-cm²/mg	No latchup observed
SEU Rate (LEO)	< 1 / year	500 km, 51.6 deg inclination
Proton Fluence	> 1e10 p/cm²	> 60 MeV protons

6. Electrical Interface

6.1 Power Architecture

The wheel has a single power input architecture. The input voltage (3.5 to 6.0 V, nominal 5 V) is regulated by an internal DC/DC converter to 3.3 V for the microcontroller and motor driver logic. The motor is driven by an H-bridge circuit with pulse-width modulation (PWM) and regenerative braking capability.

Figure 1: Power Architecture



Regenerative Braking: When the wheel decelerates under moderate torque command, the motor acts as a generator and pushes current back onto the spacecraft power bus. The spacecraft power system design must accommodate this reverse current flow. If the power line becomes disconnected during regeneration (e.g., relay switch opens), the bus voltage will rise until the overvoltage protection threshold (**6.5 V**) is reached. This causes the wheel to reset and cease regeneration.

Inrush Limiting: At power-on, the internal capacitors draw up to 0.5 A for less than 5 ms. The spacecraft power distribution unit should accommodate this transient.

6.2 Connector and Pinout

The wheel is fitted with a 6-pin Micro-D connector designated J1. The connector face view and pinout are shown below.

Figure 2: J1 Connector Face View (6-Pin Micro-D)



Pin	Signal	Wire Color	Description	Voltage Level
1	GND	Black	Power return, signal reference	0 V
2	+5V Power	Red	Main power input (3.5 to 6.0 V)	3.5 to 6.0 V
3	Tx	Orange	UART transmit (wheel to host)	3.3 V LVTTTL
4	Rx	Yellow	UART receive (host to wheel)	3.3 V LVTTTL

Pin	Signal	Wire Color	Description	Voltage Level
5	C2CK	Green	Programming/debug header (factory use only)	3.3 V
6	Shield	Bare	Chassis ground / EMI shield drain wire	Chassis

Table 1: J1 Connector Pinout (6-Pin Micro-D)

Mating connector: 6-pin Micro-D receptacle with strain relief. Recommended harness: shielded twisted-pair for Tx/Rx, separate power pair. Maximum harness length: 0.5 m recommended, 1 m maximum with shielded cable.

6.3 Power Interface Specifications

Parameter	Min	Typ	Max	Units	Notes
Operating Voltage	3.5	5.0	6.0	V	At connector pin 2
Absolute Max Voltage	-0.3		7.0	V	Exceeding causes permanent damage
Nominal Power		0.2		W	Steady state at rated momentum (0.03 Nms)
Peak Power			1.2	W	During maximum torque command (1.5 mNm)
Regenerative Current	-0.15			A	Maximum during active braking
Inrush Current			0.5	A	Duration less than 5 ms at power-on
Front-end Capacitance		470		nF	Input filter capacitor
Total Bus Capacitance		150		uF	Internal total (all rails)
OVP Threshold		6.5		V	Triggers full wheel reset
UVLO Threshold		3.2		V	Under-voltage lockout, motor disabled

6.4 Signal Interface Specifications

Parameter	Min	Typ	Max	Units	Notes
Signal Level (Tx/Rx)		3.3		V	LVTTL compatible
Absolute Max (Tx/Rx)	-0.3		3.6	V	Exceeding causes permanent damage
VOH (Tx output)	2.4		3.3	V	At IOH = -1 mA load
VOL (Tx output)	0		0.4	V	At IOL = 2 mA load
VIH (Rx input)	2.0		3.6	V	Input high threshold
VIL (Rx input)	-0.3		0.8	V	Input low threshold
Input Pullup (Rx)		10		kohm	Internal pullup to 3.3 V
ESD Protection			+/-8	kV	IEC 61000-4-2 contact discharge

7. Software Interface

7.1 Sign Convention

Positive direction is **counter-clockwise (CCW)** when viewed from the top (connector side) of the wheel assembly. A positive speed command (SET_SPEED with positive value) spins the flywheel CCW, generating a clockwise (CW) reaction torque on the spacecraft. A positive torque command (SET_TORQUE with positive value) commands CCW motor torque.

7.2 Protocol Layer 2 - Physical

The RW-0.03 communicates exclusively via asynchronous serial UART. I2C is not supported on this model (unlike the RW-0.003). The physical layer parameters are:

Parameter	Value	Notes
Interface	Asynchronous Serial UART	Full-duplex, point-to-point
Baud Rate	115200 bps	Fixed, not configurable
Data Bits	8	LSB first
Parity	None	
Stop Bits	1	
Flow Control	None	Managed by SLIP framing
Signal Level	3.3 V LVTTTL	See Section 6.4 for thresholds
Max Cable Length	0.5 m (recommended)	1 m max with shielded cable
Idle State	High (3.3 V)	Tx line idles high when no data

7.3 Protocol Layer 3 - Data Link (NSP/SLIP)

Communication uses the Nanosatellite Protocol (NSP) with SLIP (Serial Line Internet Protocol, RFC 1055) byte-stuffing for framing. SLIP encoding uses the following special byte values:

Symbol	Hex Value	Description
FEND	0xC0	Frame End delimiter, marks start and end of each frame
FESC	0xDB	Frame Escape character, precedes transposed bytes
TFEND	0xDC	Transposed Frame End, replaces 0xC0 occurring in payload
TFESC	0xDD	Transposed Frame Escape, replaces 0xDB occurring in payload

Each NSP frame is wrapped in SLIP: a FEND byte (0xC0) marks the beginning, the NSP message bytes follow with any 0xC0 or 0xDB bytes escaped, and a final FEND byte marks the end. Back-to-back frames share the FEND delimiter.

7.4 Protocol Layer 4 - Transport (NSP Message)

The NSP message format within each SLIP frame is:

Figure 5: NSP Message Frame Format

FEND 0xC0	Dest Addr 1 byte	Src Addr 1 byte	Msg Ctrl 1 byte	Data Payload 0-260 bytes	CRC-16 2 bytes	FEND 0xC0
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Field	Size (bytes)	Description
Destination Address	1	Target node address (factory-programmed, typically 0x05 for wheel)
Source Address	1	Sender node address (host typically 0x01)
Message Control	1	Bit 7: Poll/Final flag Bit 6: B (boot) flag Bit 5: ACK request Bits 4-0: Command code (0x00 to 0x1F)
Data Payload	0 to 260	Command-specific data, variable length
CRC-16	2	CRC-CCITT polynomial: $x^{16} + x^{12} + x^5 + 1$ Init value: 0xFFFF, transmitted LSB first

Maximum total message size: 265 bytes (5 header/CRC + 260 payload). All multi-byte fields use **little-endian** byte order. The CRC covers all bytes from Destination Address through the end of Data Payload.

7.5 Protocol Layer 5 - Session (Commands)

The following command codes are supported. The command code occupies bits 4-0 of the Message Control byte. Direction indicates whether the message is sent by the host (CMD) or returned by the wheel (RPL).

Code	Name	Direction	Description	Data Length
0x00	PING	CMD / RPL	Connection test. Resets watchdog timer. No payload	0 / 0
0x01	INIT	CMD	Initialize mode transition. Boot to App: data = 0x00001000 (4 bytes) App to Boot: no data (0 bytes)	0 or 4
0x02	PEEK	CMD / RPL	Read memory at specified 32-bit address. CMD sends 4 bytes (address), RPL returns data.	4 address (4 bytes)
0x03	POKE	CMD	Write data to specified 32-bit address. Payload: 4-byte address + data bytes.	variable
0x04	DIAGNOSTIC	CMD / RPL	Read diagnostic channel. CMD: channel ID (1 byte). RPL: channel data.	1 / variable
0x06	CRC	CMD / RPL	Read running firmware CRC for integrity verification.	0 / 4
0x07	READ_FILE	CMD / RPL	Read data from internal file system.	variable / variable
0x08	WRITE_FILE	CMD	Write data to internal file system.	variable
0x09	READ_EDAC	CMD / RPL	Read from EDAC-protected memory region. CMD: 4-byte address.	4 / variable
0x0A	SET_SPEED	CMD / RPL	Set wheel speed setpoint. Payload: int16 in RPM, signed. 0 = Positive = CCW.	2 / 0
0x0B	SET_TORQUE	CMD / RPL	Set torque command. Payload: int16 in mNm x 10, signed.	2 / 0
0x0C	GET_TELEMETRY	CMD / RPL	Request telemetry snapshot. Returns 18-byte packet (see Section 8).	0 / 18
0x0D	SET_MODE	CMD / RPL	Set operating mode. 0 = Idle, 1 = Speed, 2 = Torque, 3 = Test	1 / 0

Table 2: Command Code Summary

7.6 Protocol Layer 6 - Application (Modes)

The wheel operates in one of four application modes, selected via SET_MODE (0x0D):

Mode	Value	Description	Allowed Commands
Idle	0	Motor unpowered. Telemetry active. No torque output. Default for SET_MODE.	None for SET_SPEED/TORQUE
Speed Control	1	Closed-loop speed control. Wheel maintains setpoint RPM by adjusting torque.	SET_SPEED, GET_TELEMETRY, PING
Torque Control	2	Open-loop torque command. Motor applies commanded torque regardless of speed.	SET_TORQUE, GET_TELEMETRY, PING
Test	3	Factory diagnostic mode. Enables restricted commands for calibration and testing only. Not for flight use.	All commands including factory

On power-up, the wheel enters **Bootloader** mode. The host must send INIT (0x01) with 4-byte payload 0x00001000 to transition to Application mode. The wheel then defaults to **Idle** (mode 0). Transitioning between modes is instantaneous; the wheel ramps speed/torque smoothly during the transition.

8. Telemetry

8.1 Telemetry Packet Structure

The GET_TELEMETRY (0x0C) command returns an 18-byte telemetry packet with the following structure. All multi-byte fields are little-endian.

Byte Offset	Field Name	Type	Units	Range / Notes
0 to 1	wheel_speed	int16	RPM	-8000 to +8000 (positive = CCW)
2 to 3	motor_current	uint16	mA	0 to 500 (RMS phase current)
4 to 5	bus_voltage	uint16	mV	0 to 8000 (input bus voltage)
6	pcb_temperature	int8	deg C	-40 to +100 (board temperature)
7 to 8	fault_state	uint16	bitfield	See Section 10.1 for bit definitions
9	hall_state	uint8	bitfield	0x00 to 0x3F (6 Hall sensor inputs)
10 to 13	uptime	uint32	seconds	0 to 2^{32} (time since last reset)
14 to 17	cumulative_revs	uint32	revolutions	0 to 2^{32} (lifetime counter, persisted)

Table 3: Telemetry Packet Structure (18 bytes, little-endian)

8.2 Telemetry Timing

Telemetry is sampled internally at 10 Hz. The host receives a snapshot at the moment GET_TELEMETRY is processed. For continuous monitoring, the host should poll at the desired rate (recommended: 1 to 10 Hz). Each telemetry response takes approximately 2 ms to transmit at 115200 baud.

The cumulative_revs counter is stored in non-volatile EDAC memory and persists across power cycles. The uptime counter resets on each power-on or watchdog reset event.

9. Register and Memory Map

9.1 EDAC Memory Map

The wheel provides 256 bytes of Error Detection and Correction (EDAC) protected memory, addressable from 0x0000 to 0x00FF. This region is accessible via the READ_EDAC (0x09) and POKE (0x03) commands. EDAC provides single-bit error correction and double-bit error detection.

Address Range	Field Name	Type	Access	Description
0x0000 - 0x0001	speed_setpoint	int16	R/W	Current speed setpoint in RPM
0x0002 - 0x0003	torque_setpoint	int16	R/W	Current torque command in mNm x 10
0x0004 - 0x0005	speed_measured	int16	R	Measured wheel speed in RPM
0x0006 - 0x0007	current_measured	uint16	R	Motor current in mA (RMS)
0x0008 - 0x0009	bus_voltage	uint16	R	Input bus voltage in mV
0x000A	temperature	int8	R	PCB temperature in degrees C
0x000B - 0x000C	fault_state	uint16	R	Active fault bitfield (see Section 10.1)
0x000D	hall_state	uint8	R	Hall sensor state (6-bit field)
0x000E - 0x0011	cumulative_revs	uint32	R	Lifetime revolution counter (non-volatile)
0x0012 - 0x0015	operating_hours	uint32	R	Total powered-on hours (non-volatile)
0x0016 - 0x0017	reset_count	uint16	R	Total resets since factory (non-volatile)
0x0018	mode	uint8	R/W	Current operating mode (0=Idle, 1=Speed, 2=Torque, 3=Test)
0x0019 - 0x001A	speed_limit	int16	R/W	Maximum allowed speed magnitude in RPM
0x001B - 0x001C	torque_limit	int16	R/W	Maximum allowed torque magnitude in mNm x 10
0x001D - 0x001E	watchdog_timeout	uint16	R/W	Watchdog period in ms (default 500)
0x0020 - 0x003F	calibration_data	32 bytes	R	Factory calibration constants (read-only)
0x0040 - 0x005F	pid_gains	32 bytes	R/W	Speed control PID parameters (Kp, Ki, Kd, limits)
0x0060 - 0x00FF	reserved	160 bytes		Reserved for future firmware use

Table 4: EDAC Memory Map (256 bytes, 0x0000 to 0x00FF)

9.2 Diagnostic Channels

High-rate diagnostic data is accessible via the DIAGNOSTIC command (0x04). The channel ID is passed as the first byte of the command payload. Each channel returns raw ADC or sensor data at the indicated sample rate.

Channel ID	Description	Units	Sample Rate	Resolution	Notes
0	Wheel speed	RPM	100 Hz	1 RPM	Derived from Hall sensors

Channel ID	Description	Units	Sample Rate	Resolution	Notes
1	Motor current (phase A)	mA	1 kHz	0.1 mA	Per-phase measurement
2	Bus voltage	mV	100 Hz	1 mV	At DC/DC input
3	PCB temperature	deg C	1 Hz	0.5 deg C	On-board NTC thermistor
4	Reference voltage	mV	10 Hz	1 mV	Internal 3.3V rail monitor
5	Hall sensor raw	bitfield	10 kHz		6-bit raw Hall state
6	Back-EMF voltage	mV	1 kHz	1 mV	Motor phase back-EMF
7	PWM duty cycle	%	100 Hz	0.1%	Motor driver duty cycle

Table 5: Diagnostic Channel Definitions

10. Fault Handling

10.1 Fault State Bitfield

The `fault_state` register is a 16-bit field located at EDAC address 0x000B-0x000C and also reported in the telemetry packet at byte offset 7-8. Each bit indicates a specific fault condition. Bits are set by hardware or firmware detection logic. The host can clear individual fault bits by writing to the `fault_state` register via POKE, provided the underlying condition has been resolved.

Bit	Name	Description	Recovery Action
0	OVERCURRENT	Motor phase current exceeds 500 mA threshold	Auto-disables motor driver. Host must send SET_MODE to Idle, then b
1	OVERVOLTAGE	Bus voltage exceeds 6.5 V (regeneration or external) wheel reset (automatic). Enters Bootloader. Host must re-initialize.	
2	UNDERVOLTAGE	Bus voltage drops below 3.2 V UVLO threshold	Motor driver disabled. Resumes automatically when voltage recovers a
3	OVERTEMPERATURE	PCB temperature exceeds +75 deg C	Motor disabled. Resumes automatically when temperature drops below
4	HALL_SENSOR_FAULT	Invalid or stuck Hall sensor state detected	Motor disabled. Requires power cycle to clear. May indicate physical d
5	COMM_TIMEOUT	No valid command received within watchdog period (500 ms default)	When 500 ms expires, Idle mode. Any valid command clears this bit and r
6	MOTOR_STALL	Commanded speed/torque not achieved after 3 seconds	Motor driver disabled. Host must SET_MODE to Idle, then retry.
7	WATCHDOG_RESET	Internal firmware watchdog timer expired	Full reset to Bootloader mode. Host must re-initialize with INIT.
8-15	RESERVED	Reserved for future firmware revisions	

Table 6: Fault State Bitfield (uint16)

10.2 Watchdog Timer

The wheel implements a communication watchdog timer with a default timeout of **500 ms** (configurable via EDAC address 0x001D-0x001E, range 100 to 5000 ms). The host must send at least one valid command (including PING) within the timeout period to maintain active control.

If the watchdog timer expires:

- The COMM_TIMEOUT fault bit (bit 5) is set in the `fault_state` register
- The wheel transitions to Idle mode (motor unpowered, coasting to stop)
- Telemetry remains available; the wheel responds to GET_TELEMETRY requests
- Sending any valid command clears the COMM_TIMEOUT bit and restores full control
- The watchdog timer restarts on receipt of any valid command

10.3 Fault Recovery Procedures

The following table summarizes the recommended fault recovery sequence for each fault type:

Fault	Severity	Auto-Recovery	Host Action Required
OVERCURRENT	Warning	Motor auto-disabled	SET_MODE(0), wait 100ms, SET_MODE(desired), check current
OVERVOLTAGE	Critical	Full wheel reset	Wait 1s, re-init with INIT(0x00001000), reconfigure

Fault	Severity	Auto-Recovery	Host Action Required
UNDERVOLTAGE	Warning	Auto-resume at 3.5V	Monitor bus voltage; no action if transient
OVERTEMPERATURE	Warning	Auto-resume at 70C	Monitor temperature; consider duty cycle reduction
HALL_SENSOR	Critical	None	Power cycle required. If persistent, return to AnyCompany.
COMM_TIMEOUT	Warning	Idle mode	Send PING or any command to restore control
MOTOR_STALL	Warning	Motor disabled	SET_MODE(0), check for mechanical obstruction, retry
WATCHDOG	Critical	Full reset	Re-init with INIT(0x00001000), reconfigure all settings

*Table 7: Fault Recovery Procedures**End of Document*